Cyprus University of Technology



Computer Engineering & Informatics

CEI122: Ψηφιακή Λογική

CEI_122 - Report Εργαστηρίου 7

Φοιτητές:

Κάρυος Γεώργιος (30738)

Γεωργίου Ιωάννης (31292)

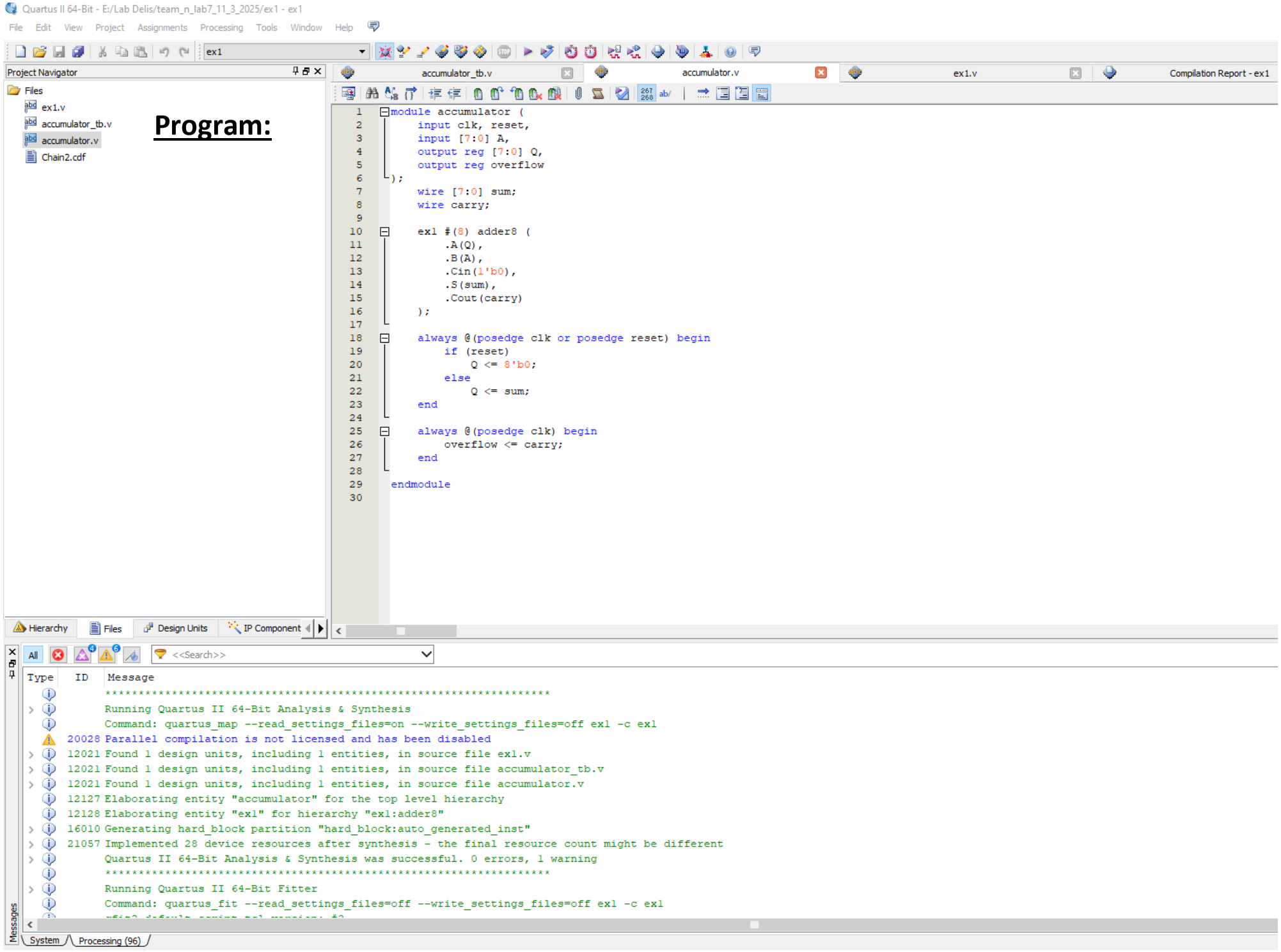
Ομάδα Ν

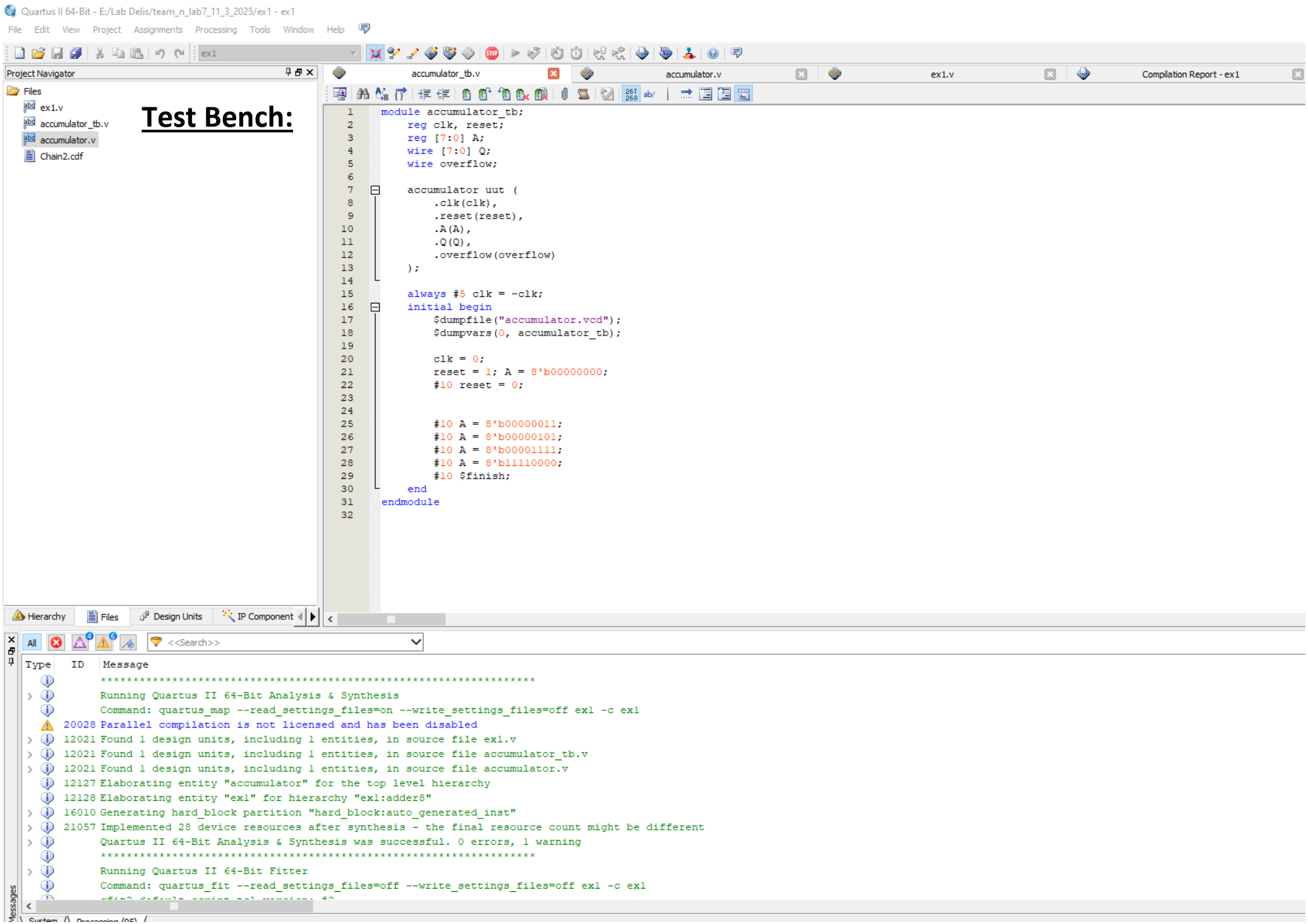
Καθηγητές:

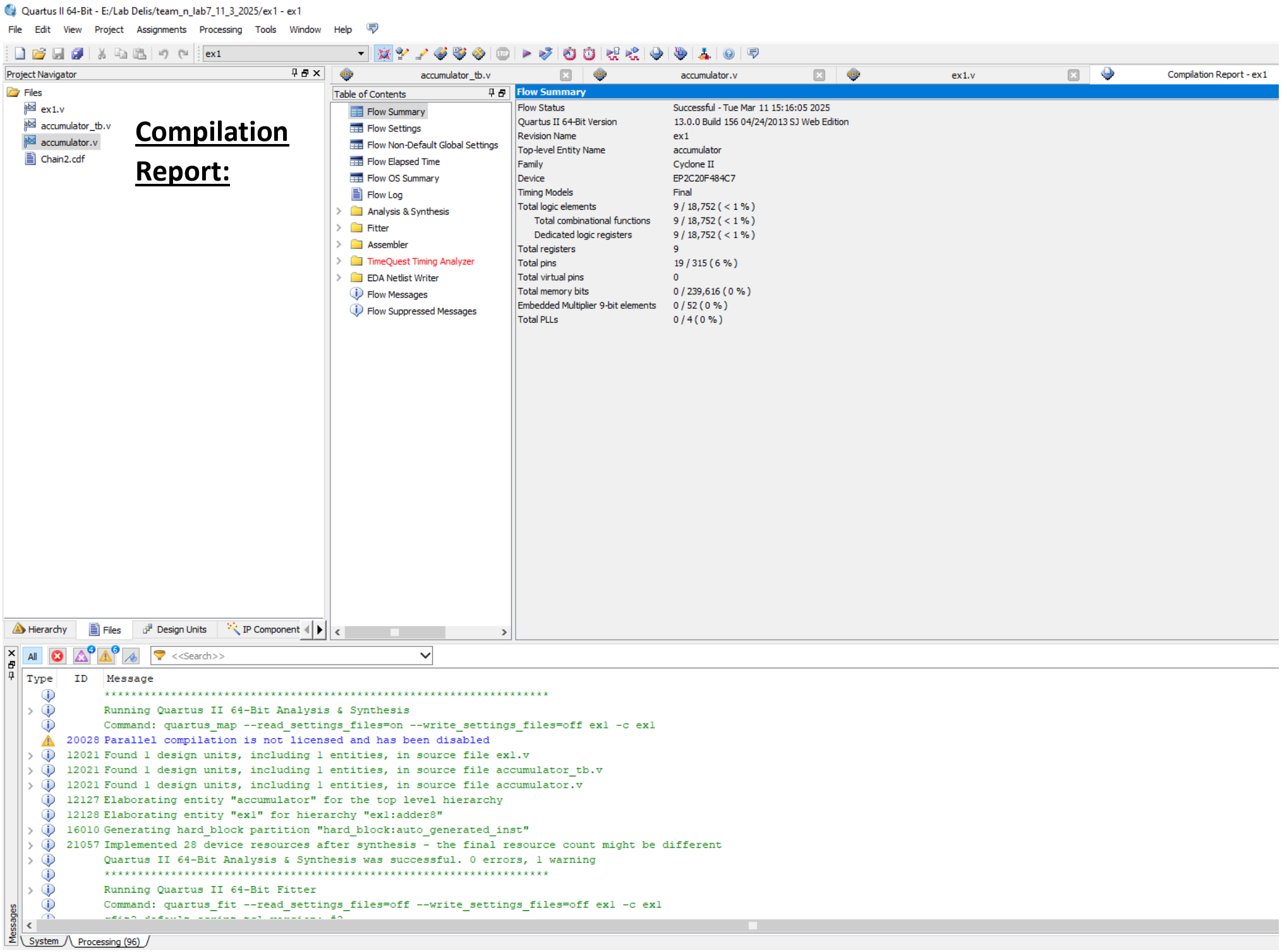
Δρ. Κυριάκος Μ. Δεληπαράσχος

κ. Χριστόδουλος Γιαννάκου

18/03/2025


Quartus II 64-Bit - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1
File Edit View Project Assignments Processing Tools Window Help
Project Navigator
Files
ex1.v
accumulator_tb.v
accumulator.v
Chain2.cdf
Program:
accumulator_tb.v
accumulator.v
ex1.v
Compilation Report - ex1
1 module accumulator (
2 input clk, reset,
3 input [7:0] A,
4 output reg [7:0] Q,
5 output reg overflow
6);
7 wire [7:0] sum;
8 wire carry;
9
10 ex1 #(8) adder8 (
11 .A(Q),
12 .B(A),
13 .Cin(1'b0),
14 .S(sum),
15 .Cout(carry)
16);
17
18 always @(posedge clk or posedge reset) begin
19 if (reset)
20 Q <= 8'b0;
21 else
22 Q <= sum;
23 end
24
25 always @(posedge clk) begin
26 overflow <= carry;
27 end
28
29 endmodule
30
Hierarchy
Files
Design Units
IP Component
All
<<Search>>
Type ID Message
Running Quartus II 64-Bit Analysis & Synthesis
Command: quartus_map --read_settings_files=on --write_settings_files=off ex1 -c ex1
20028 Parallel compilation is not licensed and has been disabled
12021 Found 1 design units, including 1 entities, in source file ex1.v
12021 Found 1 design units, including 1 entities, in source file accumulator_tb.v
12021 Found 1 design units, including 1 entities, in source file accumulator.v
12127 Elaborating entity "accumulator" for the top level hierarchy
12128 Elaborating entity "ex1" for hierarchy "ex1:adder8"
16010 Generating hard_block partition "hard_block:auto_generated_inst"
21057 Implemented 28 device resources after synthesis - the final resource count might be different
Quartus II 64-Bit Analysis & Synthesis was successful. 0 errors, 1 warning
Running Quartus II 64-Bit Fitter
Command: quartus_fit --read_settings_files=off --write_settings_files=off ex1 -c ex1
Messages
System
Processing (96)


Quartus II 64-Bit - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1
File Edit View Project Assignments Processing Tools Window Help
Project Navigator
Files
ex1.v
accumulator_tb.v
accumulator.v
Chain2.cdf
Test Bench:
accumulator_tb.v
accumulator.v
ex1.v
Compilation Report - ex1
1 module accumulator_tb;
2 reg clk, reset;
3 reg [7:0] A;
4 wire [7:0] Q;
5 wire overflow;
6
7 accumulator uut (
8 .clk(clk),
9 .reset(reset),
10 .A(A),
11 .Q(Q),
12 .overflow(overflow)
13);
14
15 always #5 clk = ~clk;
16 initial begin
17 $dumpfile("accumulator.vcd");
18 $dumpvars(0, accumulator_tb);
19
20 clk = 0;
21 reset = 1; A = 8'b00000000;
22 #10 reset = 0;
23
24
25 #10 A = 8'b00000011;
26 #10 A = 8'b00000101;
27 #10 A = 8'b00001111;
28 #10 A = 8'b11110000;
29 #10 $finish;
30 end
31 endmodule
32
Hierarchy
Files
Design Units
IP Component
Type ID Message
Running Quartus II 64-Bit Analysis & Synthesis
Command: quartus_map --read_settings_files=on --write_settings_files=off ex1 -c ex1
20028 Parallel compilation is not licensed and has been disabled
12021 Found 1 design units, including 1 entities, in source file ex1.v
12021 Found 1 design units, including 1 entities, in source file accumulator_tb.v
12021 Found 1 design units, including 1 entities, in source file accumulator.v
12127 Elaborating entity "accumulator" for the top level hierarchy
12128 Elaborating entity "ex1" for hierarchy "ex1:adder8"
16010 Generating hard_block partition "hard_block:auto_generated_inst"
21057 Implemented 28 device resources after synthesis - the final resource count might be different
Quartus II 64-Bit Analysis & Synthesis was successful. 0 errors, 1 warning
Running Quartus II 64-Bit Fitter
Command: quartus_fit --read_settings_files=off --write_settings_files=off ex1 -c ex1


Quartus II 64-Bit - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1
File Edit View Project Assignments Processing Tools Window Help
Project Navigator
Files
ex1.v
accumulator_tb.v
accumulator.v
Chain2.cdf
Compilation Report:
accumulator_tb.v
accumulator.v
ex1.v
Compilation Report - ex1
Table of Contents
Flow Summary
Flow Settings
Flow Non-Default Global Settings
Flow Elapsed Time
Flow OS Summary
Flow Log
Analysis & Synthesis
Fitter
Assembler
TimeQuest Timing Analyzer
EDA Netlist Writer
Flow Messages
Flow Suppressed Messages
Flow Summary
Flow Status Successful - Tue Mar 11 15:16:05 2025
Quartus II 64-Bit Version 13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name ex1
Top-level Entity Name accumulator
Family Cyclone II
Device EP2C20F484C7
Timing Models Final
Total logic elements 9 / 18,752 (< 1 %)
Total combinational functions 9 / 18,752 (< 1 %)
Dedicated logic registers 9 / 18,752 (< 1 %)
Total registers 9
Total pins 19 / 315 (6 %)
Total virtual pins 0
Total memory bits 0 / 239,616 (0 %)
Embedded Multiplier 9-bit elements 0 / 52 (0 %)
Total PLLs 0 / 4 (0 %)
Hierarchy
Files
Design Units
IP Component
Type ID Message
Running Quartus II 64-Bit Analysis & Synthesis
Command: quartus_map --read_settings_files=on --write_settings_files=off ex1 -c ex1
20028 Parallel compilation is not licensed and has been disabled
12021 Found 1 design units, including 1 entities, in source file ex1.v
12021 Found 1 design units, including 1 entities, in source file accumulator_tb.v
12021 Found 1 design units, including 1 entities, in source file accumulator.v
12127 Elaborating entity "accumulator" for the top level hierarchy
12128 Elaborating entity "ex1" for hierarchy "ex1:adder8"
16010 Generating hard_block partition "hard_block:auto_generated_inst"
21057 Implemented 28 device resources after synthesis - the final resource count might be different
Quartus II 64-Bit Analysis & Synthesis was successful. 0 errors, 1 warning
Running Quartus II 64-Bit Fitter
Command: quartus_fit --read_settings_files=off --write_settings_files=off ex1 -c ex1
Messages
System
Processing (96)

Quartus II 64-Bit - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

Files
- ex1.v
- accumulator_tb.v
- accumulator.v
- Chain2.cdf

accumulator_tb.v | accumulator.v | ex1.v | Compilation Report - ex1

```
1  module ex1 #(parameter N = 4) (
2      input [N-1:0] A, B,
3      input wire Cin,
4      output [N-1:0] S,
5      output Cout
6  );
7      assign {Cout, S} = A + B + Cin;
8  endmodule
9
```

Hierarchy | Files | Design Units | IP Component

Παραμετρικό 4 bit counter:

```
Type  ID     Message
      ********************************************************************
      Running Quartus II 64-Bit Analysis & Synthesis
      Command: quartus_map --read_settings_files=on --write_settings_files=off ex1 -c ex1
20028 Parallel compilation is not licensed and has been disabled
12021 Found 1 design units, including 1 entities, in source file ex1.v
12021 Found 1 design units, including 1 entities, in source file accumulator_tb.v
12021 Found 1 design units, including 1 entities, in source file accumulator.v
12127 Elaborating entity "accumulator" for the top level hierarchy
12128 Elaborating entity "ex1" for hierarchy "ex1:adder8"
16010 Generating hard_block partition "hard_block:auto_generated_inst"
21057 Implemented 28 device resources after synthesis - the final resource count might be different
      Quartus II 64-Bit Analysis & Synthesis was successful. 0 errors, 1 warning
      ********************************************************************
      Running Quartus II 64-Bit Fitter
      Command: quartus_fit --read_settings_files=off --write_settings_files=off ex1 -c ex1
```

System | Processing (96)

Altera Programmer:

Programmer - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1 - [output_files/ex1.cdf]

File Edit View Processing Tools Window Help

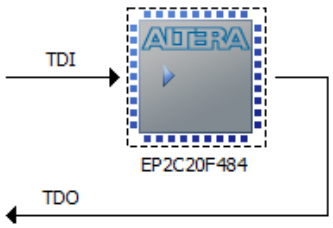
Search altera.com

Hardware Setup... | USB-Blaster [USB-0] | Mode: JTAG | Progress: 100% (Successful)

Enable real-time ISP to allow background programming (for MAX II and MAX V devices)

Start | Stop | Auto Detect | Delete | Add File... | Change File... | Save File | Add Device... | Up | Down

File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/ex1.sof	EP2C20F484	001B24C7	001B24C7	☑	☐	☐	☐



Pin Planner - E:/Lab Delis/team_n_lab7_11_3_2025/ex1 - ex1

File Edit View Processing Tools Window Help

Groups

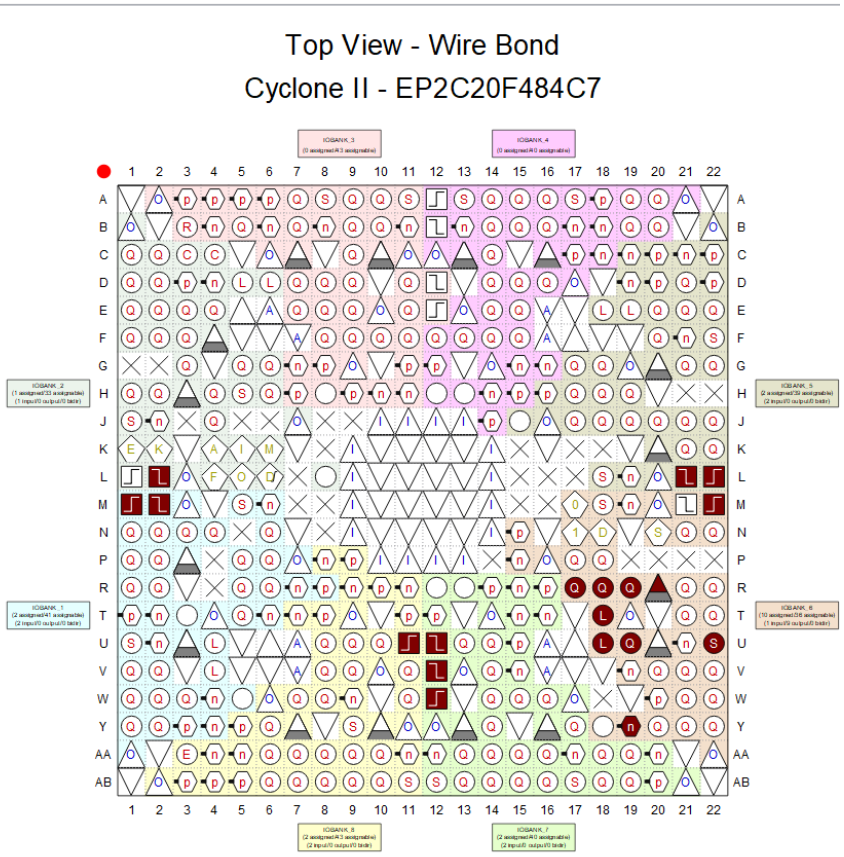
Named: *

Node Name	Direction
A[7..0]	Input Group
Q[7..0]	Output Group
<<new group>>	

Groups Report

Tasks

- Run Analysis and Elaboration
- Early Pin Planning
  - Early Pin Planning...
  - Run I/O Assignment Analys
  - Export Pin Assignments...
- Change View
  - Show I/O Banks
  - Show VREF Groups
  - Show Edges
  - Show DQ/DQS Pins
    - x4 Mode



Named: * Edit:

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Strength	Differential Pair
A[7]	Input	PIN_L22	5	B5_N1	PIN_L22	3.3-V LV...default)		24mA (default)	
A[6]	Input	PIN_L21	5	B5_N1	PIN_L21	3.3-V LV...default)		24mA (default)	
A[5]	Input	PIN_M22	6	B6_N0	PIN_M22	3.3-V LV...default)		24mA (default)	
A[4]	Input	PIN_V12	7	B7_N1	PIN_V12	3.3-V LV...default)		24mA (default)	
A[3]	Input	PIN_W12	7	B7_N1	PIN_W12	3.3-V LV...default)		24mA (default)	
A[2]	Input	PIN_U12	8	B8_N0	PIN_U12	3.3-V LV...default)		24mA (default)	
A[1]	Input	PIN_U11	8	B8_N0	PIN_U11	3.3-V LV...default)		24mA (default)	
A[0]	Input	PIN_M2	1	B1_N0	PIN_M2	3.3-V LV...default)		24mA (default)	
Q[7]	Output	PIN_R20	6	B6_N0	PIN_R20	3.3-V LV...default)		24mA (default)	
Q[6]	Output	PIN_R19	6	B6_N0	PIN_R19	3.3-V LV...default)		24mA (default)	
Q[5]	Output	PIN_U19	6	B6_N1	PIN_U19	3.3-V LV...default)		24mA (default)	
Q[4]	Output	PIN_Y19	6	B6_N1	PIN_Y19	3.3-V LV...default)		24mA (default)	

All Pins

